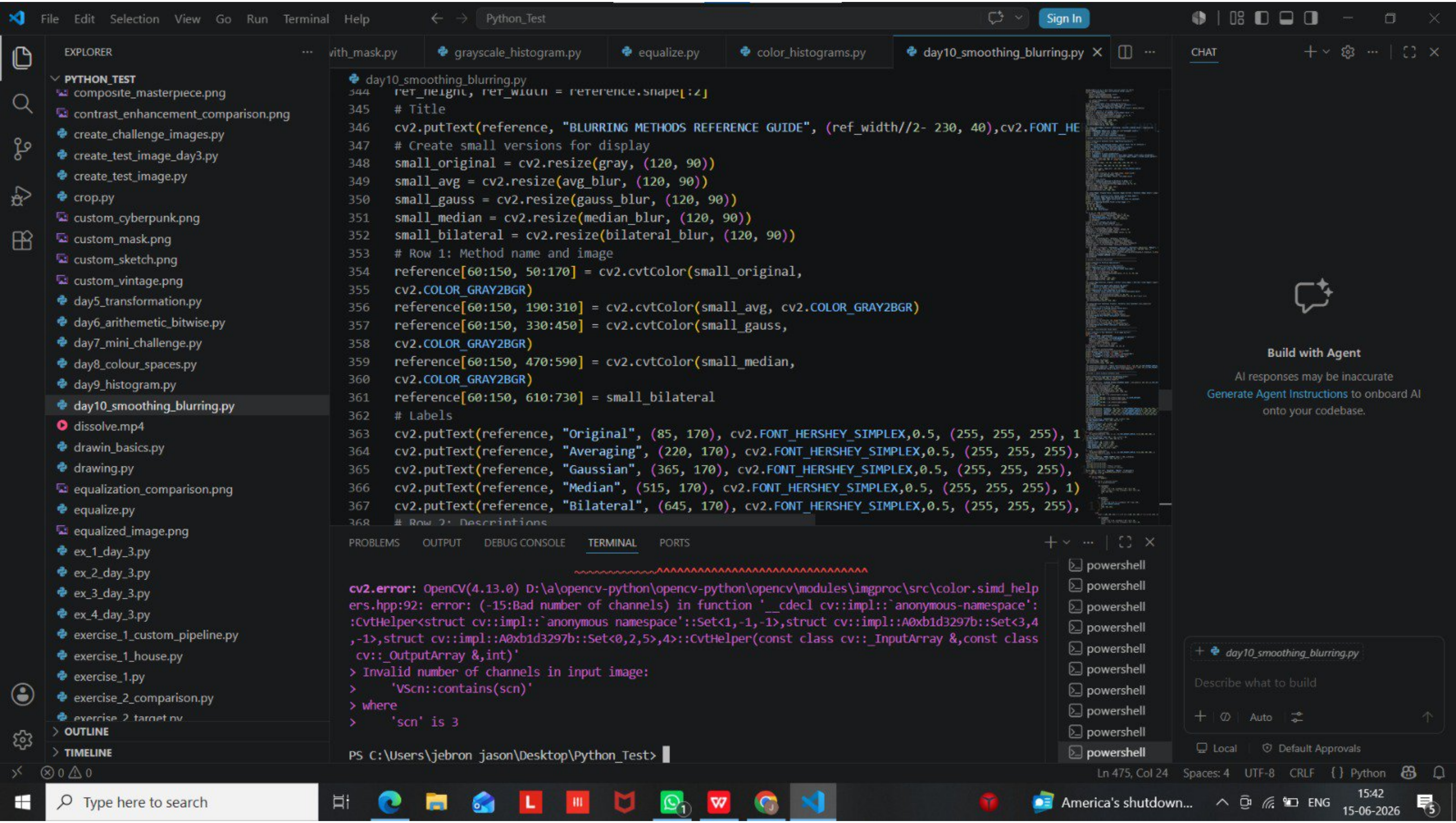
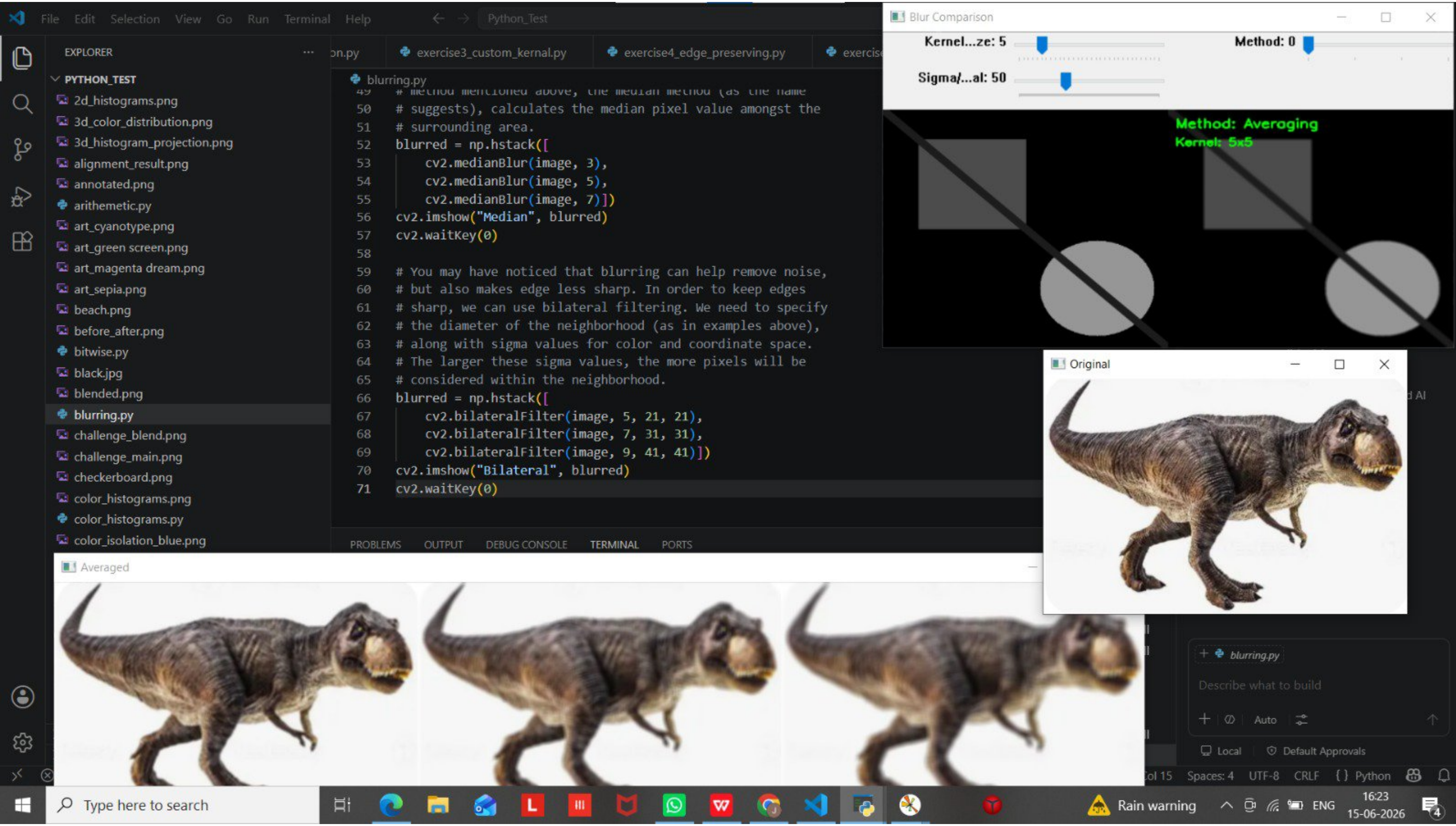
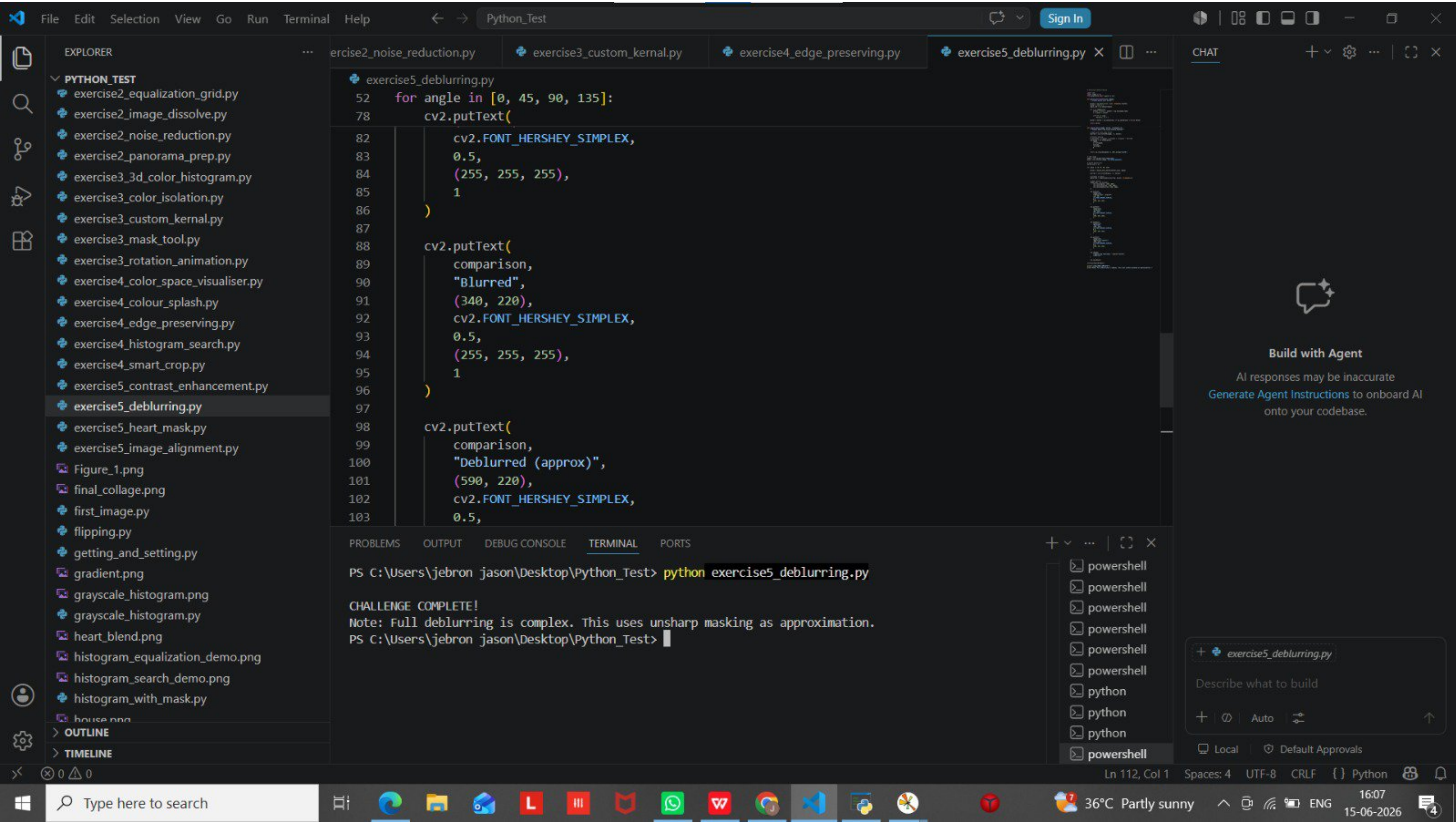


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
composite_masterpiece.png
contrast_enhancement_comparison.png
create_challenge_images.py
create_test_image_day3.py
create_test_image.py
crop.py
custom_cyberpunk.png
custom_mask.png
custom_sketch.png
custom_vintage.png
day5_transformation.py
day6_arithemetic_bitwise.py
day7_mini_challenge.py
day8_colour_spaces.py
day9_histogram.py
day10_smoothing_blurring.py
dissolve.mp4
drawin_basics.py
drawing.py
equalization_comparison.png
equalize.py
equalized_image.png
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_custom_pipeline.py
exercise_1_house.py
exercise_1.py
exercise_2_comparison.py
OUTLINE
TIMELINE
grayscale_histogram.py
equalize.py
color_histograms.py
day10_smoothing_blurring.py
345 # Title
346 cv2.putText(reference, "BLURRING METHODS REFERENCE GUIDE", (ref_width//2- 230, 40),cv2.FONT_HE
347 # Create small versions for display
348 small_original = cv2.resize(gray, (120, 90))
349 small_avg = cv2.resize(avg_blur, (120, 90))
350 small_gauss = cv2.resize(gauss_blur, (120, 90))
351 small_median = cv2.resize(median_blur, (120, 90))
352 small_bilateral = cv2.resize(bilateral_blur, (120, 90))
353 # Row 1: Method name and image
354 reference[60:150, 50:170] = cv2.cvtColor(small_original,
355 cv2.COLOR_GRAY2BGR)
356 reference[60:150, 190:310] = cv2.cvtColor(small_avg, cv2.COLOR_GRAY2BGR)
357 reference[60:150, 330:450] = cv2.cvtColor(small_gauss,
358 cv2.COLOR_GRAY2BGR)
359 reference[60:150, 470:590] = cv2.cvtColor(small_median,
360 cv2.COLOR_GRAY2BGR)
361 reference[60:150, 610:730] = small_bilateral
362 # Labels
363 cv2.putText(reference, "Original", (85, 170), cv2.FONT_HERSHEY_SIMPLEX,0.5, (255, 255, 255), 1
364 cv2.putText(reference, "Averaging", (220, 170), cv2.FONT_HERSHEY_SIMPLEX,0.5, (255, 255, 255),
365 cv2.putText(reference, "Gaussian", (365, 170), cv2.FONT_HERSHEY_SIMPLEX,0.5, (255, 255, 255),
366 cv2.putText(reference, "Median", (515, 170), cv2.FONT_HERSHEY_SIMPLEX,0.5, (255, 255, 255), 1)
367 cv2.putText(reference, "Bilateral", (645, 170), cv2.FONT_HERSHEY_SIMPLEX,0.5, (255, 255, 255),
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
cv2.error: OpenCV(4.13.0) D:\a\opencv-python\opencv-python\opencv\modules\imgproc\src\color.simd_helpers.hpp:92: error: (-15:Bad number of channels) in function '__cdecl cv::impl::`anonymous-namespace'::CvtHelper<struct cv::impl::`anonymous namespace'::Set<1,-1,-1>,struct cv::impl::A0xb1d3297b::Set<3,4,-1>,struct cv::impl::A0xb1d3297b::Set<0,2,5>,4>::CvtHelper(const class cv::_InputArray &,const class cv::_OutputArray &,int)'
> Invalid number of channels in input image:
> 'VScn::contains(scn)'
> where
> 'scn' is 3
PS C:\Users\jebron jason\Desktop\Python_Test>
powershell
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
day10_smoothing_blurring.py
Describe what to build
Auto
Local
Default Approvals
Ln 475, Col 24 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
America's shutdown...
ENG
15:42
15-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
EXPLORER
PYTHON_TEST
2d_histograms.png
3d_color_distribution.png
3d_histogram_projection.png
alignment_result.png
annotated.png
arithemetic.py
art_cyanotype.png
art_green screen.png
art_magenta dream.png
art_sepia.png
beach.png
before_after.png
bitwise.py
black.jpg
blended.png
blurring.py
challenge_blend.png
challenge_main.png
checkerboard.png
color_histograms.png
color_histograms.py
color_isolation_blue.png
exercise3_custom_kernal.py
exercise4_edge_preserving.py
blurring.py
# suggests), calculates the median pixel value amongst the
# surrounding area.
blurred = np.hstack([
cv2.medianBlur(image, 3),
cv2.medianBlur(image, 5),
cv2.medianBlur(image, 7)])
cv2.imshow("Median", blurred)
cv2.waitKey(0)
# You may have noticed that blurring can help remove noise,
# but also makes edge less sharp. In order to keep edges
# sharp, we can use bilateral filtering. We need to specify
# the diameter of the neighborhood (as in examples above),
# along with sigma values for color and coordinate space.
# The larger these sigma values, the more pixels will be
# considered within the neighborhood.
blurred = np.hstack([
cv2.bilateralFilter(image, 5, 21, 21),
cv2.bilateralFilter(image, 7, 31, 31),
cv2.bilateralFilter(image, 9, 41, 41)])
cv2.imshow("Bilateral", blurred)
cv2.waitKey(0)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Blur Comparison
Kernel...ze: 5
Method: 0
Sigma/...al: 50
Method: Averaging
Kernel: 5x5
Original
Averaged
blurring.py
Describe what to build
Auto
Local
Default Approvals
Spaces: 4 UTF-8 CRLF {} Python
Type here to search
Rain warning
ENG
16:23
15-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise2_equalization_grid.py
exercise2_image_dissolve.py
exercise2_noise_reduction.py
exercise2_panorama_prep.py
exercise3_3d_color_histogram.py
exercise3_color_isolation.py
exercise3_custom_kernal.py
exercise3_mask_tool.py
exercise3_rotation_animation.py
exercise4_color_space_visualiser.py
exercise4_colour_splash.py
exercise4_edge_preserving.py
exercise4_histogram_search.py
exercise4_smart_crop.py
exercise5_contrast_enhancement.py
exercise5_deblurring.py
exercise5_heart_mask.py
exercise5_image_alignment.py
Figure_1.png
final_collage.png
first_image.py
flipping.py
getting_and_setting.py
gradient.png
grayscale_histogram.png
grayscale_histogram.py
heart_blend.png
histogram_equalization_demo.png
histogram_search_demo.png
histogram_with_mask.py
OUTLINE
TIMELINE
ercise2_noise_reduction.py
exercise3_custom_kernal.py
exercise4_edge_preserving.py
exercise5_deblurring.py
exercise5_deblurring.py
for angle in [0, 45, 90, 135]:
cv2.putText(
cv2.FONT_HERSHEY_SIMPLEX,
0.5,
(255, 255, 255),
1
)
cv2.putText(
comparison,
"Blurred",
(340, 220),
cv2.FONT_HERSHEY_SIMPLEX,
0.5,
(255, 255, 255),
1
)
cv2.putText(
comparison,
"Deblurred (approx)",
(590, 220),
cv2.FONT_HERSHEY_SIMPLEX,
0.5,
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise5_deblurring.py
CHALLENGE COMPLETE!
Note: Full deblurring is complex. This uses unsharp masking as approximation.
PS C:\Users\jebron jason\Desktop\Python_Test>
powershell
python
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
exercise5_deblurring.py
Describe what to build
Auto
Local
Default Approvals
Ln 112, Col 1 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
36°C Partly sunny
ENG
16:07
15-06-2026

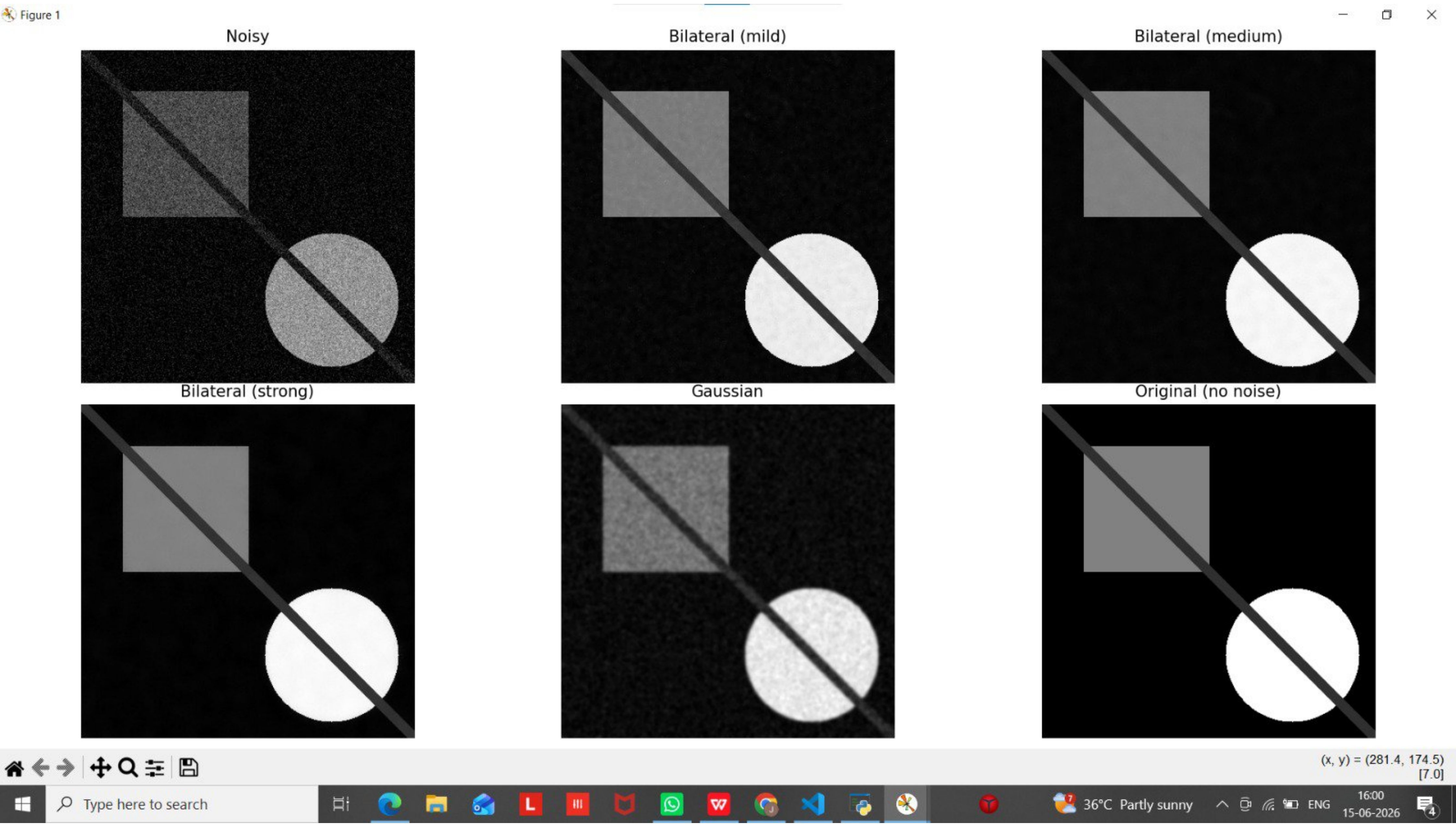
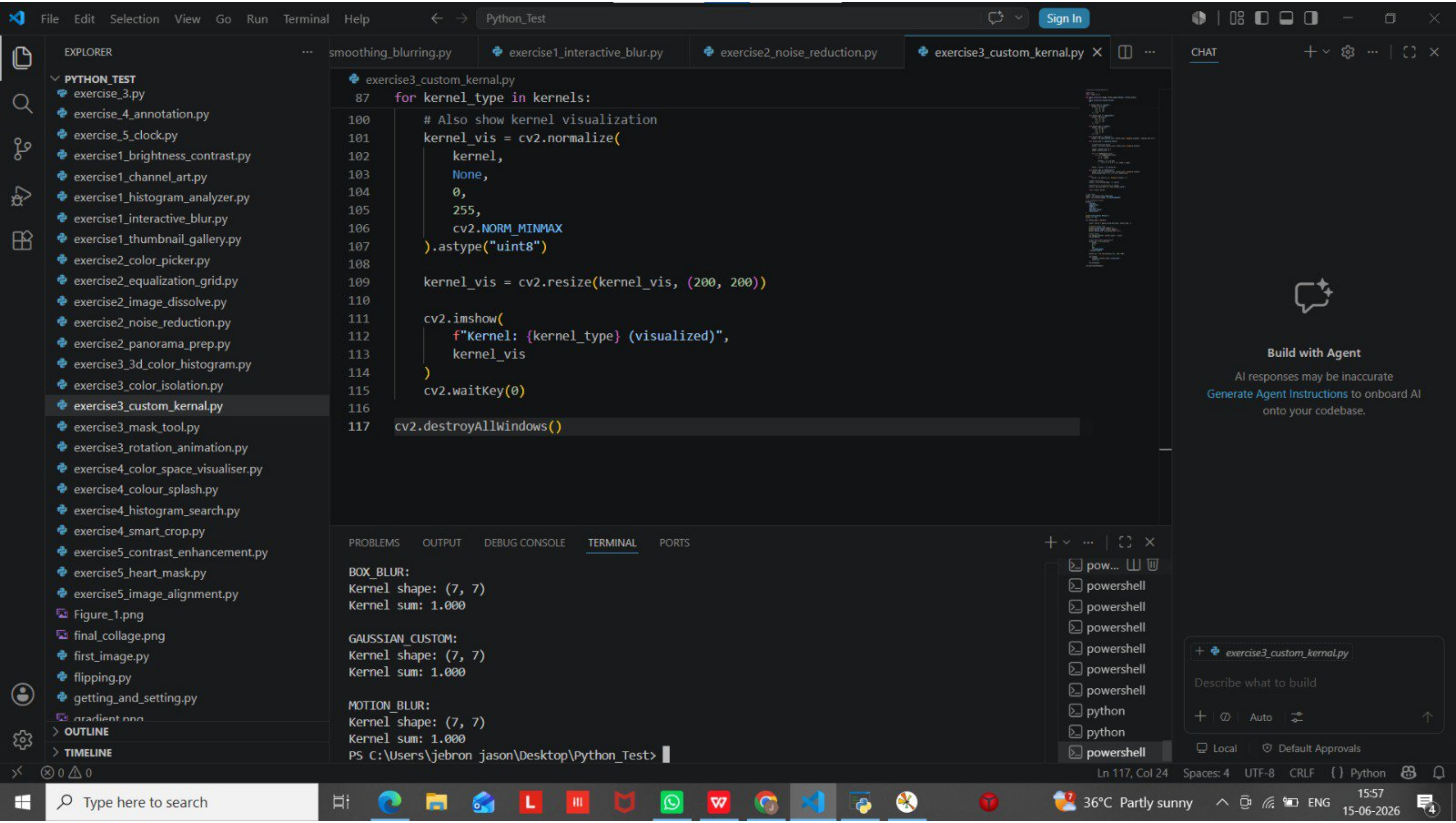


Figure 1
Noisy
Bilateral (mild)
Bilateral (medium)
Bilateral (strong)
Gaussian
Original (no noise)
(x, y) = (281.4, 174.5)
[7.0]
Type here to search
36°C Partly sunny
ENG
16:00
15-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise_3.py
exercise_4_annotation.py
exercise_5_clock.py
exercise1_brightness_contrast.py
exercise1_channel_art.py
exercise1_histogram_analyzer.py
exercise1_interactive_blur.py
exercise1_thumbnail_gallery.py
exercise2_color_picker.py
exercise2_equalization_grid.py
exercise2_image_dissolve.py
exercise2_noise_reduction.py
exercise2_panorama_prep.py
exercise3_3d_color_histogram.py
exercise3_color_isolation.py
exercise3_custom_kernal.py
exercise3_mask_tool.py
exercise3_rotation_animation.py
exercise4_color_space_visualiser.py
exercise4_colour_splash.py
exercise4_histogram_search.py
exercise4_smart_crop.py
exercise5_contrast_enhancement.py
exercise5_heart_mask.py
exercise5_image_alignment.py
Figure_1.png
final_collage.png
first_image.py
flipping.py
getting_and_setting.py
OUTLINE
TIMELINE
smoothing_blurring.py
exercise1_interactive_blur.py
exercise2_noise_reduction.py
exercise3_custom_kernal.py
for kernel_type in kernels:
# Also show kernel visualization
kernel_vis = cv2.normalize(
kernel,
None,
0,
255,
cv2.NORM_MINMAX
).astype("uint8")
kernel_vis = cv2.resize(kernel_vis, (200, 200))
cv2.imshow(
f"Kernel: {kernel_type} (visualized)",
kernel_vis
)
cv2.waitKey(0)
cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
BOX_BLUR:
Kernel shape: (7, 7)
Kernel sum: 1.000
GAUSSIAN_CUSTOM:
Kernel shape: (7, 7)
Kernel sum: 1.000
MOTION_BLUR:
Kernel shape: (7, 7)
Kernel sum: 1.000
PS C:\Users\jebron jason\Desktop\Python_Test>
powershell
python
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
Ln 117, Col 24 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
36°C Partly sunny
ENG
15:57
15-06-2026

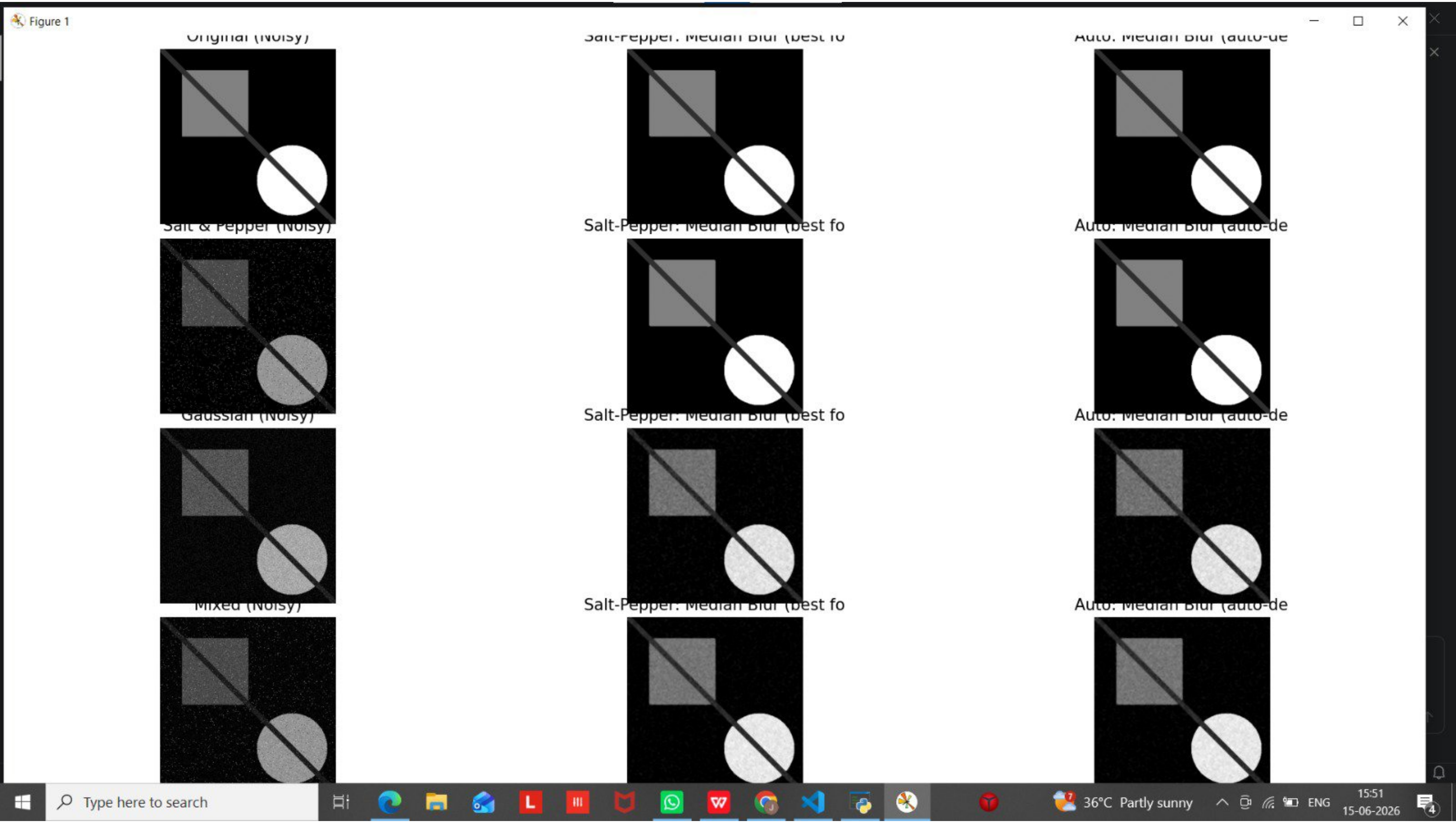
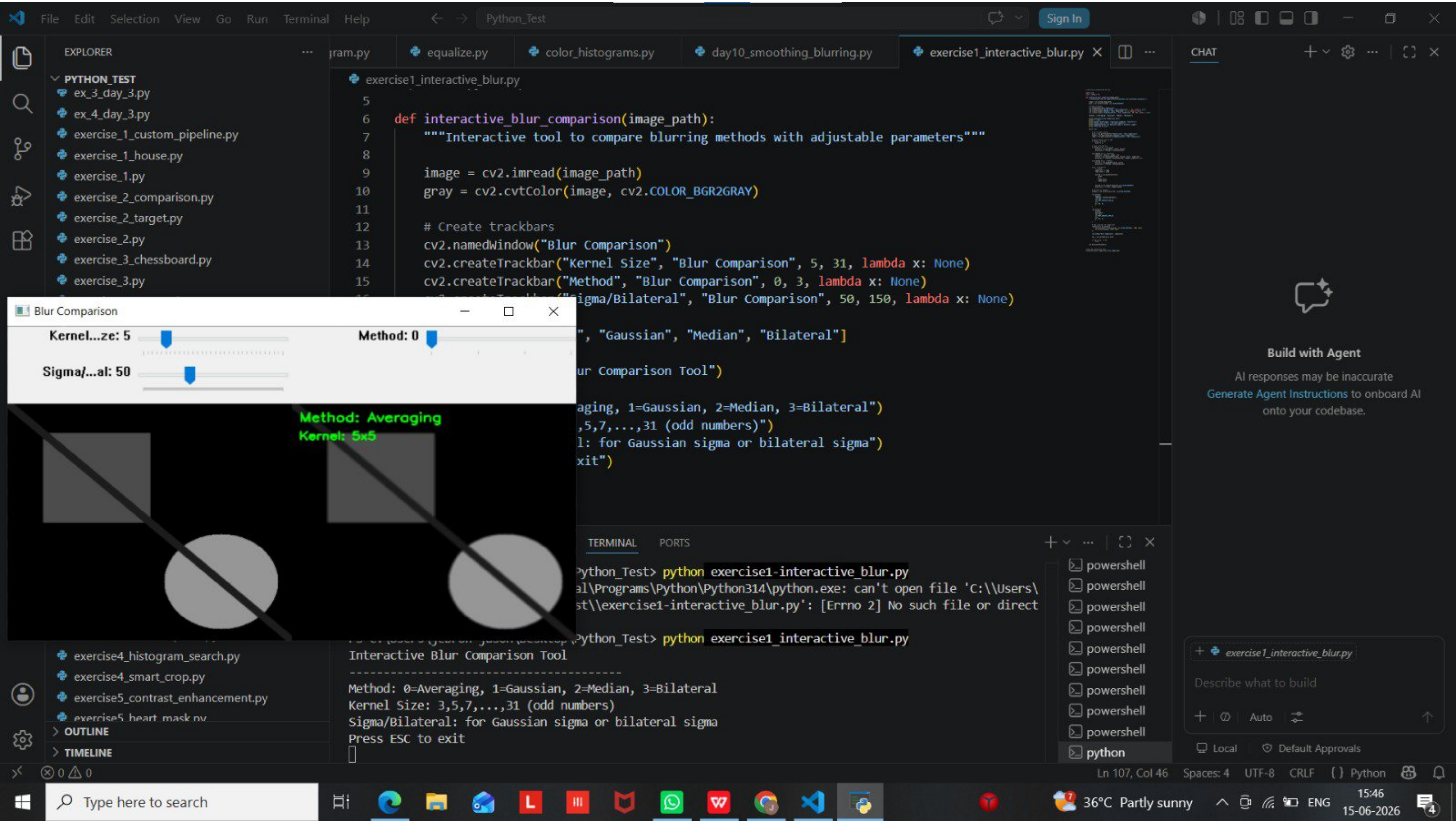


Figure 1
Salt-Pepper: Median Blur (best fo
Auto: Median Blur (auto-de
Salt-Pepper: Median Blur (best fo
Auto: Median Blur (auto-de
Salt-Pepper: Median Blur (best fo
Auto: Median Blur (auto-de
Type here to search
36°C Partly sunny
ENG
15:51
15-06-2026



exercise1_interactive_blur.py
def interactive_blur_comparison(image_path):
"""Interactive tool to compare blurring methods with adjustable parameters"""
image = cv2.imread(image_path)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Create trackbars
cv2.namedWindow("Blur Comparison")
cv2.createTrackbar("Kernel Size", "Blur Comparison", 5, 31, lambda x: None)
cv2.createTrackbar("Method", "Blur Comparison", 0, 3, lambda x: None)
Blur Comparison
Kernel...ze: 5
Method: 0
Sigma/...al: 50
Method: Averaging
Kernel: 5x5
TERMINAL
PORTS
Interactive Blur Comparison Tool
Method: 0=Averaging, 1=Gaussian, 2=Median, 3=Bilateral
Kernel Size: 3,5,7,...,31 (odd numbers)
Sigma/Bilateral: for Gaussian sigma or bilateral sigma
Press ESC to exit
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build